

German-born Maria Goeppert Mayer was a theoretical physicist who, in 1963, became the second woman to win the Nobel Prize in Physics. She created the nuclear shell model, which explained how some isotopes (variants of an element) would be more stable than others and why some were likely to be radioactive.

Maria Goeppert Mayer was born in Kattowitz (now in Poland) and studied mathematics and then physics at the University of Göttingen. The university was a leading center of the new field of quantum mechanics, and Goeppert Mayer studied under renowned physicist Max Born. In her doctoral thesis in 1930, Goeppert Mayer predicted that some atoms could absorb two photons (the smallest unit of light) simultaneously, and as a result move to a higher energy state. Her theory was not proven until 1961, when the invention of the laser made it possible to demonstrate the effect in a crystal of the metal europium. Multiphoton absorption has since been used for body scanning and creating polymers (chemicals made of repeating units).

Goeppert Mayer moved to the US in 1930, when her husband accepted a position as assistant professor of chemistry at Johns Hopkins University, and she spent many years as an unpaid researcher at Johns Hopkins and Columbia Universities. During this time, she published a number of scientific papers, including one, in 1935, on a type of radioactive decay called double-beta decay. In 1942, she was invited

MILESTONES

CHANGE OF DIRECTION

Changes course to study physics after attending an inspirational quantum mechanics seminar in 1924.

PAID POST

After years of unpaid work, is offered a senior role in 1946 at Argonne National Laboratory, Illinois.

CATALYST QUESTION

A colleague's question prompts her nuclear shell model; publishes her theory in *Physical Review* in 1950.

NOBEL PRIZE

In 1963, is jointly awarded the Nobel Prize in Physics for her discoveries on nuclear shell structure.

During the Manhattan Project, Goeppert Mayer joined Edward Teller's team at the Los Alamos Laboratory, New Mexico, to assist with the development of the first hydrogen bomb: a powerful bomb that used nuclear fusion and fission.



“Physics is **puzzle solving** ... but of puzzles **created by nature**, not by the mind of man.”

Maria Goeppert Mayer